[738] (CONCLUDED)
NOTE ON A HYPERGEOMETRIC SERIES.

viz. this is

$$= -\frac{(\alpha - \alpha')\, X}{1 + X + X^2}.$$

We have

$$\begin{aligned} QP' + PQ' &= \alpha'^2\,(-\alpha'' + \alpha X) + \alpha^2\,(\alpha - \alpha' X), \\ &= \alpha^3 - \alpha'^3 - (\alpha' - \alpha'')\, X, \quad = (\alpha - \alpha')\, X; \end{aligned}$$

and

$$PQ = -\alpha' + (\alpha' + \alpha'')\, X - \alpha' X^2, \quad = 1 + X + X^2;$$

hence

$$\frac{1}{PQ}(QP' + PQ') = \frac{(\alpha - \alpha')\, X}{1 + X + X^2},$$

and the sum of the two parts is $= 0$.

Similarly as regards the second equation, the second part

$$\frac{3X}{1 - X^3}\, PQ\,(PQ' + P'Q) - \frac{3X}{1 - X^3}\, P^2 Q^2$$

is

$$= \frac{3PQX}{1 - X^3}\,\{(PQ' + P'Q)X - PQ\}.$$

Here $PQ' + P'Q$ is $\alpha\,(\alpha - \alpha' X) - \alpha'^2(-\alpha' + \alpha X)$, which is $= 1 + 2X$; and PQ being $= 1 + X + X^2$, the term in $\{\ \}$ is

$$(1 + 2X)\, X - (1 + X + X^2), \quad = -(1 - X)(1 + X);$$

hence, outside the $\{\ \}$ writing for PQ its value $= 1 + X + X^2$, the term is

$$= \frac{-3X(1 + X + X^2)(1 - X)(1 + X)}{1 - X^3}, \quad = -3X(1 + X),$$

which is the value of the second part in question; the first part is

$$(PQ' + QP')^2 - PQ\, P'Q' \quad = (1 + 2X)^2 - (1 + X + X^2), \quad = 3X(1 + X);$$

and the sum of the two terms is thus $= 0$.

739.

NOTE ON THE OCTAHEDRON FUNCTION.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xvi. (1879), pp. 280, 281.]

A SEXTIC function

$$U = (a,\, b,\, c,\, d,\, e,\, f,\, g \mid x,\, y)^6,$$

such that its fourth derivative

$$\begin{aligned} (U,\, U)_4 = \quad & (ae - 4bd + 3c^2)\, x^4 \\ & + 2\,(af - 3be + 2cd)\, x^3 y \\ & + (ag - 9ce + 8d^2)\, x^2 y^2 \\ & + 2\,(bg - 3cf + 2de)\, xy^3 \\ & + (cg - 4df + 3e^2)\, y^4 \end{aligned}$$

is identically $= 0$, is considered by Dr Klein, and is called by him the octahedron function. Supposing that by a linear transformation the function is made to contain the factors x, y, or what is the same thing assuming $a = 0$, $g = 0$, then the equations to be satisfied become

$$-4bd + 3c^2 = 0, \quad -3be + 2cd = 0, \quad -9ce + 8d^2 = 0, \quad -3cf + 2de = 0, \quad -4df + 3e^2 = 0,$$

which are all satisfied if only $c = d = e = 0$; and then assuming, as is allowable,

$$b = -f = 1,$$

we have his canonical form $xy(x^4 - y^4)$ of the octahedron function.

But the equations may be satisfied in a different manner; viz. the first and last equations give

$$b = \frac{3c^2}{4d}, \quad f = \frac{3e^2}{4d},$$

and, substituting these in the remaining equations, they become

$$\frac{c}{4d}(-9ce + 8d^2) = 0, \quad -9ce + 8d^2 = 0, \quad \frac{e}{4d}(-9ce + 8d^2) = 0,$$

all satisfied if only $-9ce + 8d^2 = 0$. Assuming $b = f = 2$, the values are

$$b,\ c,\ d,\ e,\ f \ = \ 2,\ 2\sqrt{2},\ 3,\ 2\sqrt{2},\ 2,$$

and the form is

$$\begin{aligned} &\left(2,\ 2\sqrt{2},\ 3,\ 2\sqrt{2},\ 2 \mid x,\ y\right)^6. \\ &= xy\left(x^2 + \tfrac{3}{\sqrt{2}}\,xy + 2y^2\right)\left(x^2 + \sqrt{2}\,xy + y^2\right) \\ &= xy\left(x + \tfrac{1+i}{\sqrt{2}}\,y\right)\left(x + \tfrac{1-i}{\sqrt{2}}\,y\right)\left(x + \sqrt{2}\,y\right)\left(x + \tfrac{y}{\sqrt{2}}\right). \end{aligned}$$

This is, in fact, a linear transformation of the foregoing form $XY(X^4 - Y^4)$; for writing

$$X = \left(x + \tfrac{1+i}{\sqrt{2}}\,y\right), \qquad Y = \left(x + \tfrac{1-i}{\sqrt{2}}\,y\right),$$

we have

$$\begin{aligned} X^2 &= x^2 + (1+i)\sqrt{2}\,xy + iy^2, \\ Y^2 &= x^2 + (1-i)\sqrt{2}\,xy - iy^2; \end{aligned}$$

and therefore

$$\begin{aligned} X^2 + Y^2 &= 2\{x + \sqrt{2}\,y\}, \\ X^2 - Y^2 &= 2i\sqrt{2}\,y\left(x + \tfrac{y}{\sqrt{2}}\right), \end{aligned}$$

or finally

$$XY(X^4 - Y^4) = 4i\sqrt{2}\,xy\ \left(x + \tfrac{1+i}{\sqrt{2}}\,y\right)\left(x + \tfrac{1-i}{\sqrt{2}}\,y\right)\left(x + \sqrt{2}\,y\right)\left(x + \tfrac{y}{\sqrt{2}}\right),$$

and the two forms are thus identical.

740.

ON CERTAIN ALGEBRAICAL IDENTITIES.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xvi. (1879), pp. 281, 282.]

If P_0, P_1, P_2 are points on a circle, say the circle $x^2 + y^2 = 1$, then it is possible to find functions of (P_0, P_1) and of (P_1, P_2) respectively, which are really independent of P_1, and consequently functions of only P_0 and P_2: the expression "function of a point or points" being here used to mean algebraical function of the coordinates of the point or points. Thus the functions of (P_0, P_1) and of (P_1, P_2) being $x_0x_1 + y_0y_1$, $x_0y_1 - x_1y_0$, and $x_1x_2 + y_1y_2$, $x_1y_2 - x_2y_1$, we have

$$(x_0x_1 + y_0y_1)(x_1x_2 + y_1y_2) + (x_0y_1 - x_1y_0)(x_1y_2 - x_2y_1) = x_0x_2 + y_0y_2,$$

and another like equation. This depends obviously on the circumstance that the coordinates of a point of the circle are expressible by means of the functions sin, cos, $x = \cos u$, $y = \sin u$ and the identity written down is obtained by expressing the cosine of $u_2 - u_0$, $= (u_2 - u_1) + (u_1 - u_0)$, in terms of the cosines and sines of $u_2 - u_1$ and $u_1 - u_0$.

Evidently the like property holds good for a curve, such that the coordinates of any point of it can be expressed by means of "additive" functions of a parameter u; where, by an additive function $f(u)$, is meant a function such that $f(u+v)$ is an algebraical function of $f(u)$, $f(v)$: the sine and cosine are each of them an additive function, because

$$\sin(u+v) = \sin u\sqrt{1-\sin^2 v} + \sin v\sqrt{1-\sin^2 u},$$

and, similarly, for the cosine. But it is convenient to consider pairs or groups $f(u)$, $\phi(u)$, ..., where $f(u+v)$, $\phi(u+v)$, ... are each of them an algebraical (rational) function of $f(u)$, $\phi(u)$, ..., $f(v)$, $\phi(v)$, ...; the sine and cosine are such a group, and so also are the elliptic functions sn, cn, dn; but the H and Θ, or say the ϑ-functions generally, are not additive.

In the case of the elliptic functions, we may consider the quadriquadric curve

$$x^2 + y^2 = 1, \qquad k^2x^2 + z^2 = 1,$$

so that the coordinates of a point on the curve are $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$. Taking then P_0, P_1, P_2, points on the curve, and (x_0, y_0, z_0), (x_1, y_1, z_1), (x_2, y_2, z_2), the coordinates of these points respectively, we have in the same way, from $u_2 - u_0 = (u_2 - u_1) + (u_1 - u_0)$, three equations, of which the first is

$$\frac{x_0y_2z_2 - x_2y_0z_0}{1 - k^2x_0^2x_2^2} = \frac{(1-k^2x_1^2x_2^2)(x_0y_1z_1 - x_1y_0z_0)(y_1y_2 + x_1z_1x_2z_2)(z_1z_2 + k^2x_1y_1x_2y_2) + (1-k^2x_0^2x_1^2)(x_1y_2z_2 - x_2y_1z_1)(y_0y_1 + x_0z_0x_1z_1)(z_0z_1 + k^2x_0y_0x_1y_1)}{(1-k^2x_0^2x_1^2)^2(1-k^2x_1^2x_2^2)^2 - k^2(x_0y_1z_1 - x_1y_0z_0)^2(x_1y_2z_2 - x_2y_1z_1)^2}.$$

The form of the right-hand side is

$$\frac{A + Bx_1^2y_1^2z_1^2}{C + Dx_1^2y_1^2z_1^2}$$

where A, B, C, D are each of them rational as regards x_1^2; and it is easy to see that the equation can only subsist under the condition that we have separately

$$\frac{x_0y_2z_2 - x_2y_0z_0}{1 - k^2x_0^2x_2^2} = \frac{A}{C} = \frac{B}{D},$$

implying of course the identity $AD - BC = 0$. The values of B and D are found without difficulty; we, in fact, have

$$\begin{aligned} B &= 2k^2\,(x_2^2 - x_0^2)\,(x_0y_0z_0 + x_2y_2z_2), \\ D &= 2k^2\,(x_0y_2z_2 + x_2y_0z_0)(x_0^2y_2^2z_2^2 + x_2^2y_0^2z_0^2), \end{aligned}$$

so that, comparing the left-hand side with $B \div D$, we have the identity

$$x_0^2y_2^2z_2^2 - x_2^2y_0^2z_0^2 = (x_2^2 - x_0^2)(1 - k^2x_0^2x_2^2),$$

which is right. The comparison with $A \div C$ would be somewhat more difficult to effect.

741.

ON A THEOREM OF ABEL'S RELATING TO A QUINTIC EQUATION.

[From the Proceedings of the Cambridge Philosophical Society, vol. iii. (1880), pp. 155–159.]

The theorem in question is given, Œuvres Complètes, [Christiania, 1881], t. ii., p. 266, as an extract from a letter to Crelle dated 14th March, 1826, as follows:

"Si une équation du cinquième degré dont les coefficients sont des nombres rationnels est résoluble algébriquement, on peut donner aux racines la forme suivante

$$x = c + Aa^{\frac{1}{5}}a_1^{\frac{2}{5}}a_2^{\frac{3}{5}}a_3^{\frac{4}{5}} + A_1 a_1^{\frac{1}{5}}a_2^{\frac{2}{5}}a_3^{\frac{3}{5}}a^{\frac{4}{5}} + A_2 a_2^{\frac{1}{5}}a_3^{\frac{2}{5}}a^{\frac{3}{5}}a_1^{\frac{4}{5}} + A_3 a_3^{\frac{1}{5}}a^{\frac{2}{5}}a_1^{\frac{3}{5}}a_2^{\frac{4}{5}},$$

où

$$\begin{aligned}
a &= m + n\sqrt{1+e^2} + \sqrt{h(1+e^2+\sqrt{1+e^2})},\\
a_1 &= m - n\sqrt{1+e^2} + \sqrt{h(1+e^2-\sqrt{1+e^2})},\\
a_2 &= m + n\sqrt{1+e^2} - \sqrt{h(1+e^2+\sqrt{1+e^2})},\\
a_3 &= m - n\sqrt{1+e^2} - \sqrt{h(1+e^2-\sqrt{1+e^2})},\\
A &= K + K'a + K''a_2 + K'''aa_2, \quad A_1 = K + K'a_1 + K''a_3 + K'''a_1a_3,\\
A_2 &= K + K'a_2 + K''a + K'''a_2a, \quad A_3 = K + K'a_3 + K''a_1 + K'''a_3a_1.
\end{aligned}$$

Les quantités c, h, e, m, n, K, K', K'', K''' sont des nombres rationnels. Mais de cette manière l'équation $x^5 + ax + b = 0$ n'est pas résoluble tant que a et b sont des quantités quelconques. J'ai trouvé de pareils théorèmes pour les équations du $7^{\text{ième}}$, $11^{\text{ième}}$, $13^{\text{ième}}$, etc. degré."

It is easy to see that x is the root of a quintic equation, the coefficients of which are rational and integral functions of a, a_1, a_2, a_3: these coefficients are not symmetrical functions of a, a_1, a_2, a_3, but they are functions which remain unaltered by the cyclical change a into a_1, a_1 into a_2, a_2 into a_3, a_3 into a. But the coefficients of the quintic equation must be rational functions of c, h, e, m, n, K, K', K'', K'''; hence regarding a, a_1, a_2, a_3, as the roots of a quartic equation, the coefficients of this equation being rational functions of m, n, e, h, this equation must be such that every rational function of the roots, unchanged by the aforesaid cyclical change of the roots, shall be rationally expressible in terms of these quantities m, n, e, h: or, what is the same thing, the group of the quartic equation, using the term "group of the equation" in the sense assigned to it by Galois, must be $aa_1a_2a_3$, $a_1a_2a_3a$, $a_2a_3aa_1$, $a_3aa_1a_2$. And conversely, the quartic equation being of this form, x will be the root of a quintic equation, the coefficients whereof are rational and integral functions of c, h, e, m, n, K, K', K'', K'''.

To investigate the form of a quartic equation having the property just referred to, let it be proposed to find γ, γ' functions of e, h, such that $\gamma^2 + \gamma'^2$ is a rational function of e, h, but that $\gamma^2 - \gamma'^2$, $\gamma\gamma'$ are rational multiples of the same quadric radical $\sqrt{\theta}$. Assume that we have

$$\gamma^2 = p + q\sqrt{\theta}, \qquad \gamma'^2 = p - q\sqrt{\theta};$$

then

$$(\gamma^2 + \gamma'^2)^2 = 4(p^2 + q^2)\,\theta;$$

that $\gamma^2 + \gamma'^2$ may be rational, we must have $p^2 + q^2 = \lambda^2\theta$, or say $p^2 + q^2 = h^2\theta$; hence, $\theta = \dfrac{p^2}{h^2} + \dfrac{q^2}{h^2}$ must be a sum of two squares, or, assuming one of these equal to unity and the other of them equal to e^2, say $\theta = 1 + e^2$, we satisfy the required equation by taking $p = h$, $q = he$: viz. we thus have

$$\gamma^2 - \gamma'^2 = 2h\sqrt{1+e^2}, \quad \gamma\gamma' = he\sqrt{1+e^2}, \quad \gamma^2 + \gamma'^2 = 2h(1+e^2),$$

and thence also

$$\gamma^2 = h(1+e^2+\sqrt{1+e^2}), \quad \gamma'^2 = h(1+e^2-\sqrt{1+e^2}),$$

the roots of these expressions, or values of γ, γ', being such that

$$\gamma\gamma' = he\sqrt{1+e^2}.$$

Taking now α rational, $= m$ suppose, and β a rational multiple of

$$\sqrt{1+e^2}, \quad = n\sqrt{1+e^2},$$

suppose; it is easy to see that the quartic equation which has for its roots

$$a,\ a_1,\ a_2,\ a_3\ =\ \alpha+\beta+\gamma,\quad \alpha-\beta+\gamma',\quad \alpha+\beta-\gamma,\quad \alpha-\beta-\gamma',$$

has the property in question, viz. that every rational function of the roots unchangeable by the cyclical change a into a_1, a_1 into a_2, a_2 into a_3, a_3 into a, is rationally expressible in terms of e, h, m, n.

It will be sufficient to give the proof in the case of a rational and integral function; such a function, unchangeable as aforesaid, is of the form

$$\phi(a,\ a_1,\ a_2,\ a_3)+\phi(a_1,\ a_2,\ a_3,\ a)+\phi(a_2,\ a_3,\ a,\ a_1)+\phi(a_3,\ a,\ a_1,\ a_2);$$

and if $\phi(a,\ a_1,\ a_2,\ a_3)$ contains a term $\alpha^m\beta^n\gamma^p\gamma'^q$, then the other three functions will contain respectively the terms

$$\alpha^m(-\beta)^n\gamma'^p(-\gamma)^q,\quad \alpha^m\beta^n(-\gamma)^p(-\gamma')^q,\quad \alpha^m(-\beta)^n(-\gamma')^p\gamma^q;$$

viz. the sum of the four terms is

$$=\alpha^m\beta^n\left[\{1+(-)^{p+q}\}\,\gamma^p\gamma'^q+\{(-)^n+(-)^{n+p}\}\,\gamma'^p\gamma^q\right].$$

This obviously vanishes unless p and q are both even, or both odd; and the cases to be considered are 1°, n even, p and q even; 2°, n odd, p and q even; 3°, n even, p and q odd; 4°, n odd, p and q odd. Writing, for greater distinctness, $2n$ or $2n+1$ for n, according as n is even or odd, and similarly for p and q, the term is, in the four cases respectively,

$$\begin{aligned}
&=2\alpha^m\beta^{2n}\,(\gamma^{2p}\gamma'^{2q}+\gamma^{2q}\gamma'^{2p}),\\
&=2\alpha^m\beta^{2n+1}\,(\gamma^{2p}\gamma'^{2q}-\gamma^{2q}\gamma'^{2p}),\\
&=2\alpha^m\beta^{2n}\,\gamma\gamma'\,(\gamma^{2p}\gamma'^{2q}+\gamma^{2q}\gamma'^{2p}),\\
&=2\alpha^m\beta^{2n+1}\,\gamma\gamma'\,(\gamma^{2p}\gamma'^{2q}-\gamma^{2q}\gamma'^{2p}).
\end{aligned}$$

The second, third, and fourth expressions contain the factors

$$\beta\,(\gamma^2-\gamma'^2),\quad \gamma\gamma'\,(\gamma^2-\gamma'^2),\quad \beta\gamma\gamma',$$

respectively; and the first expression as it stands, and the other three divested of these factors respectively are rational functions of α, β^2, γ^2, γ'^2, that is, they are rational functions of m, n, e, h. But the omitted factors $\beta(\gamma^2-\gamma'^2)$, $\gamma\gamma'(\gamma^2-\gamma'^2)$, $\beta\gamma\gamma'$, $=2nh(1+e^2)$, $2h^2e(1+e^2)$, $nhe(1+e^2)$ are rational functions of n, h, e; hence each of the original four expressions is a rational function of m, n, h, e; and the entire function

$$\phi(a,\ a_1,\ a_2,\ a_3)+\phi(a_1,\ a_2,\ a_3,\ a)+\phi(a_2,\ a_3,\ a,\ a_1)+\phi(a_3,\ a,\ a_1,\ a_2)$$

is a rational function of m, n, h, e.

Replacing α, β, γ, γ' by their values, the roots of the quartic equation are

$$\begin{aligned}
&m+n\sqrt{1+e^2}+\sqrt{h(1+e^2+\sqrt{1+e^2})},\\
&m-n\sqrt{1+e^2}+\sqrt{h(1+e^2-\sqrt{1+e^2})},\\
&m+n\sqrt{1+e^2}-\sqrt{h(1+e^2+\sqrt{1+e^2})},\\
&m-n\sqrt{1+e^2}-\sqrt{h(1+e^2-\sqrt{1+e^2})}.
\end{aligned}$$

And I stop to remark that taking $m,\ n,\ e,\ h=-\frac{1}{4},\ +\frac{1}{4},\ 2,\ -\frac{5}{16}$ respectively, the roots are

$$\begin{aligned}
&-\tfrac{1}{4}+\tfrac{1}{4}\sqrt{5}+\sqrt{-\tfrac{1}{8}(5+\sqrt{5})},\\
&-\tfrac{1}{4}-\tfrac{1}{4}\sqrt{5}+\sqrt{-\tfrac{1}{8}(5-\sqrt{5})},\\
&-\tfrac{1}{4}+\tfrac{1}{4}\sqrt{5}-\sqrt{-\tfrac{1}{8}(5+\sqrt{5})},\\
&-\tfrac{1}{4}-\tfrac{1}{4}\sqrt{5}-\sqrt{-\tfrac{1}{8}(5-\sqrt{5})};
\end{aligned}$$

viz. these are the imaginary fifth roots of unity, or roots r, r^2, r^3, r^4 of the quartic equation $x^4+x^3+x^2+x+1=0$; which equation, as is well known, has the group $rr^2r^3r^4$, $r^2r^4rr^3$, $r^3rr^4r^2$, $r^4r^3r^2r$.

Reverting to Abel's expression for x, and writing this for a moment in the form

$$x = c+p+s+r+q,$$

the quintic equation in x is

$$\begin{aligned} 0 = {} & (x-c)^5 \\ & + (x-c)^3 \cdot -5\,(pr+qs) \\ & + (x-c)^2 \cdot -5\,(p^2s+q^2p+r^2s+s^2r) \\ & + (x-c) \quad \cdot -5\,(p^2q+q^2r+r^2s+s^2p)+5\,(p^2r^2+q^2s^2)-5pqrs \\ & + (x-c)^0 \cdot -(p^5+q^5+r^5+s^5) \\ & \qquad\qquad +5\,(p^3rs+q^3sp+r^3pq+s^3qr) \\ & \qquad\qquad -5\,(p^2q^2s+q^2r^2p+r^2s^2q+s^2p^2r). \end{aligned}$$

If we substitute herein for p, q, r, s their values, then, altering the order of the terms, the final result is found to be

$$\begin{aligned} 0 = {} & (x-c)^5 \\ & + (x-c)^3 \cdot -5\,(AA_1+A_2A_3)\,aa_1a_2a_3 \\ & + (x-c)^2 \cdot -5\,(A^2A_1a_1a_2a_3+A_1^2A_2a_2a_3a+A_2^2A_3a_3aa_1+A_3^2Aaa_1a_2)\,aa_1a_2a_3 \\ & + (x-c) \quad \cdot -5\,(A^2A_2a_1a_3+A_1^2A_3a_2a+A_2^2Aa_3a_1+A_3^2A_1aa_2)\,aa_1a_2a_3 \\ & \qquad\qquad +5\,(A^2A_2^2+A_1^2A_3^2-AA_1A_2A_3)\,(aa_1a_2a_3)^2 \\ & + (x-c)^0 \cdot -(A^5a_1a_2^2a_3^3+A_1^5a_2a_3^2a^3+A_2^5a_3a^2a_1^3+A_3^5aa_1^2a_2^3)\,aa_1a_2a_3 \\ & \qquad\qquad +5\,(A^3A_2A_3a_1a_2a_3+A_1^3A_3Aa_2a_3a+A_2^3AA_1a_3aa_1+A_3^3A_1A_2aa_1a_2)\,(aa_1a_2a_3)^2 \\ & \qquad\qquad -5\,(A^3A_1^2A_2a_1a_3+A_1^3A_2^2A_3a_2a+A_2^3A_3^2Aa_3a_1+A_3^3A^2A_1aa_2)\,(aa_1a_2a_3)^2, \end{aligned}$$

viz. considering herein A, A_1, A_2, A_3 as standing for their values

$$K+K'a+K''a_2+K'''aa_2, \quad \text{\&c.}$$

respectively, each coefficient is a function of a, a_1, a_2, a_3, which is unaltered by the cyclical change of these values and therefore is a rational function of

$$m,\ n,\ e,\ h,\ K,\ K',\ K'',\ K'''.$$

742.

ON THE TRANSFORMATION OF COORDINATES.

[From the Proceedings of the Cambridge Philosophical Society, vol. iii. (1880), pp. 178–184.]

The formulae for the transformation between two sets of oblique coordinates assume a very elegant form when presented in the notation of matrices. I call to mind that a matrix denotes a system of quantities arranged in a square form

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix};$$

see my "Memoir on the Theory of Matrices," *Phil. Trans.* t. CXLVIII. (1858), pp. 17–37, [152]; moreover $(\alpha,\, \beta,\, \gamma \mid x,\, y,\, z)$ denotes $\alpha x + \beta y + \gamma z$, and so

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} (x,\, y,\, z)$$

denotes

$$(\alpha x + \beta y + \gamma z,\ \alpha' x + \beta' y + \gamma' z,\ \alpha'' x + \beta'' y + \gamma'' z),$$

and again

$$(\alpha,\, \beta,\, \gamma \mid x,\, y,\, z \mid \xi,\, \eta,\, \zeta) \text{ denotes } \xi(\alpha x + \beta y + \gamma z) + \eta(\alpha' x + \beta' y + \gamma' z) + \zeta(\alpha'' x + \beta'' y + \gamma'' z).$$

Consequently

$$(\alpha,\, \beta,\, \gamma \mid x,\, y,\, z \mid \xi,\, \eta,\, \zeta) = (\alpha,\, \alpha',\, \alpha'' \mid \xi,\, \eta,\, \zeta \mid x,\, y,\, z).$$

In the case of a symmetrical matrix

$$\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix},$$

the equal expressions

$$(a,\, h,\, g \mid x,\, y,\, z \mid \xi,\, \eta,\, \zeta) = (a,\, h,\, g \mid \xi,\, \eta,\, \zeta \mid x,\, y,\, z)$$

are also written

$$(a,\, b,\, c,\, f,\, g,\, h \mid x,\, y,\, z \mid \xi,\, \eta,\, \zeta), \text{ or } (a,\, \ldots \mid \xi,\, \eta,\, \zeta \mid x,\, y,\, z).$$

In particular, if $x = \xi$, $y = \eta$, $z = \zeta$, then

$$(a,\, h,\, g \mid x,\, y,\, z)^2 \text{ is written } (a,\, b,\, c,\, f,\, g,\, h \mid x,\, y,\, z)^2.$$

Two matrices are compounded together according to the law

$$\begin{pmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{pmatrix} \begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} = \begin{pmatrix} (a,\, b,\, c \mid \alpha,\, \alpha',\, \alpha'') & (a,\, b,\, c \mid \beta,\, \beta',\, \beta'') & (a,\, b,\, c \mid \gamma,\, \gamma',\, \gamma'') \\ (a',\, b',\, c' \mid \alpha,\, \alpha',\, \alpha'') & (a',\, b',\, c' \mid \beta,\, \beta',\, \beta'') & (a',\, b',\, c' \mid \gamma,\, \gamma',\, \gamma'') \\ (a'',\, b'',\, c'' \mid \alpha,\, \alpha',\, \alpha'') & (a'',\, b'',\, c'' \mid \beta,\, \beta',\, \beta'') & (a'',\, b'',\, c'' \mid \gamma,\, \gamma',\, \gamma'') \end{pmatrix},$$

viz. in the compound matrix, the top-line is

$$(a,\, b,\, c \mid \alpha,\, \alpha',\, \alpha''),\ \ (a,\, b,\, c \mid \beta,\, \beta',\, \beta''),\ \ (a,\, b,\, c \mid \gamma,\, \gamma',\, \gamma''),$$

and the other two lines are the like functions with $(a',\, b',\, c')$, and $(a'',\, b'',\, c'')$, respectively, in the place of $(a,\, b,\, c)$.

The inverse matrix is the matrix the terms of which are the minors of the determinant formed out of the original matrix, each minor being divided by this determinant, viz.

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix}^{-1} = \frac{1}{\nabla} \begin{pmatrix} \beta'\gamma'' - \beta''\gamma' & \beta''\gamma - \beta\gamma'' & \beta\gamma' - \beta'\gamma \\ \gamma'\alpha'' - \gamma''\alpha' & \gamma''\alpha - \gamma\alpha'' & \gamma\alpha' - \gamma'\alpha \\ \alpha'\beta'' - \alpha''\beta' & \alpha''\beta - \alpha\beta'' & \alpha\beta' - \alpha'\beta \end{pmatrix},$$

where ∇ is the determinant

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix}.$$

Coming now to the question of transformation, write

	x	y	z
x	1	ν	μ
y	ν	1	λ
z	μ	λ	1

	x_1	y_1	z_1
x	α	α'	α''
y	β	β'	β''
z	γ	γ'	γ''

	x_1	y_1	z_1
x_1	1	ν_1	μ_1
y_1	ν_1	1	λ_1
z_1	μ_1	λ_1	1

	x	y	z
x_1	α	β	γ
y_1	α'	β'	γ'
z_1	α''	β''	γ''

viz. the axes of x, y, z are inclined to each other at angles the cosines whereof are λ, μ, ν, those of x_1, y_1, z_1 are inclined to each other at angles the cosines whereof are λ_1, μ_1, ν_1, and the cosines of the inclinations of the two sets of axes to each other are α, β, γ; α', β', γ'; α'', β'', γ'': as is more clearly indicated in the diagram, the top-line showing that cosine-inclinations of x to

$$x,\ y,\ z,\ x_1,\ y_1,\ z_1,$$

are

$$1,\ \nu,\ \mu,\ \alpha,\ \alpha',\ \alpha'',$$

respectively, and the like for the other lines of the diagram. The letters Ω, Ω_1, V, W are used to denote matrices, viz. as appearing by the diagram, these are

$$\Omega = \begin{pmatrix} 1 & \nu & \mu \\ \nu & 1 & \lambda \\ \mu & \lambda & 1 \end{pmatrix}, \quad \Omega_1 = \begin{pmatrix} 1 & \nu_1 & \mu_1 \\ \nu_1 & 1 & \lambda_1 \\ \mu_1 & \lambda_1 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix}, \quad W = \begin{pmatrix} \alpha & \alpha' & \alpha'' \\ \beta & \beta' & \beta'' \\ \gamma & \gamma' & \gamma'' \end{pmatrix},$$

respectively.

The coordinates $(x,\ y,\ z)$ and $(x_1,\ y_1,\ z_1)$ form each set a broken line extending from the origin to the point; hence projecting on the axes of x, y, z and on those of x_1, y_1, z_1 respectively, we have two sets, each of three equations, which may be written

$$(\Omega \mid x,\ y,\ z) = (W \mid x_1,\ y_1,\ z_1), \qquad (V \mid x,\ y,\ z) = (\Omega_1 \mid x_1,\ y_1,\ z_1);$$

where of course each set implies the other set.

We have

$$\begin{aligned} (x,\ y,\ z) &= (\Omega^{-1}W \mid x_1,\ y_1,\ z_1), \quad = (V^{-1}\Omega_1 \mid x_1,\ y_1,\ z_1), \\ (x_1,\ y_1,\ z_1) &= (W^{-1}\Omega \mid x,\ y,\ z), \quad = (\Omega_1^{-1}V \mid x,\ y,\ z), \end{aligned}$$

the first giving in two forms $(x,\ y,\ z)$ as linear functions of $(x_1,\ y_1,\ z_1)$, and the second giving in two forms $(x_1,\ y_1,\ z_1)$ as linear functions of $(x,\ y,\ z)$; comparing the two forms for each set, we have

$$\Omega^{-1}W = V^{-1}\Omega_1, \qquad W^{-1}\Omega = \Omega_1^{-1}V,$$

or, what is the same thing,

$$V\Omega^{-1}V = \Omega_1, \qquad W\Omega_1^{-1}W = \Omega,$$

where in each equation the two sides are matrices which must be equal term by term to each other; but, the matrices being symmetrical, the equation thus gives (not nine but only) six equations. Writing

$$(a,\ b,\ c,\ f,\ g,\ h) = (1-\lambda^2,\ 1-\mu^2,\ 1-\nu^2,\ \mu\nu-\lambda,\ \nu\lambda-\mu,\ \lambda\mu-\nu),$$

and

$$K = 1 - \lambda^2 - \mu^2 - \nu^2 + 2\lambda\mu\nu,$$

we have

$$\Omega^{-1} = \frac{1}{K}\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}.$$

The first equation, written in the form

$$V\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}V = K\Omega_1,$$

denotes the six equations

$$\begin{aligned}
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma)^2 &= K, \\
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha',\, \beta',\, \gamma')^2 &= K, \\
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha'',\, \beta'',\, \gamma'')^2 &= K, \\
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha',\, \beta',\, \gamma' \mid \alpha'',\, \beta'',\, \gamma'') &= K\lambda_1, \\
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha'',\, \beta'',\, \gamma'' \mid \alpha,\, \beta,\, \gamma) &= K\mu_1, \\
(a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma \mid \alpha',\, \beta',\, \gamma') &= K\nu_1.
\end{aligned}$$

And, similarly, writing

$$(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1) = (1-\lambda_1^2,\ 1-\mu_1^2,\ 1-\nu_1^2,\ \mu_1\nu_1-\lambda_1,\ \nu_1\lambda_1-\mu_1,\ \lambda_1\mu_1-\nu_1),$$

and

$$K_1 = 1-\lambda_1^2-\mu_1^2-\nu_1^2+2\lambda_1\mu_1\nu_1,$$

then

$$\Omega_1^{-1} = \frac{1}{K_1}\begin{pmatrix} a_1 & h_1 & g_1 \\ h_1 & b_1 & f_1 \\ g_1 & f_1 & c_1 \end{pmatrix};$$

and the second equation, written in the form

$$W\begin{pmatrix} a_1 & h_1 & g_1 \\ h_1 & b_1 & f_1 \\ g_1 & f_1 & c_1 \end{pmatrix}V = K_1\Omega,$$

denotes the six equations

$$\begin{aligned}
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \alpha,\, \alpha',\, \alpha'')^2 &= K_1, \\
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \beta,\, \beta',\, \beta'')^2 &= K_1, \\
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \gamma,\, \gamma',\, \gamma'')^2 &= K_1, \\
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \beta,\, \beta',\, \beta'' \mid \gamma,\, \gamma',\, \gamma'') &= K_1\lambda, \\
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \gamma,\, \gamma',\, \gamma'' \mid \alpha,\, \alpha',\, \alpha'') &= K_1\mu, \\
(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1 \mid \alpha,\, \alpha',\, \alpha'' \mid \beta,\, \beta',\, \beta'') &= K_1\nu.
\end{aligned}$$

The two sets each of six equations are, in fact, equivalent to a single set of six equations, and serve to express the relations between the nine cosines

$$(\alpha,\, \beta,\, \gamma,\, \alpha',\, \beta',\, \gamma',\, \alpha'',\, \beta'',\, \gamma'')$$

and the cosines $(\lambda,\, \mu,\, \nu)$ and $(\lambda_1,\, \mu_1,\, \nu_1)$. Observe that the nine cosines are not (as in the rectangular transformation) the coefficients of transformation between the two sets of coordinates.

From the original linear relations between the coordinates, multiplying the equations of the first set by $x,\, y,\, z$ and adding, and again multiplying the equations of the second set by $(x_1,\, y_1,\, z_1)$ and adding, we have

$$\begin{aligned}
(\Omega \mid x,\, y,\, z)^2 &= (W \mid x_1,\, y_1,\, z_1 \mid x,\, y,\, z), \\
(\Omega_1 \mid x_1,\, y_1,\, z_1)^2 &= (V \mid x,\, y,\, z \mid x_1,\, y_1,\, z_1).
\end{aligned}$$

But

$$(W \mid x_1,\, y_1,\, z_1 \mid x,\, y,\, z)$$

and

$$(V \mid x,\, y,\, z \mid x_1,\, y_1,\, z_1)$$

denote one and the same function; hence

$$(\Omega \mid x,\, y,\, z)^2 = (\Omega_1 \mid x_1,\, y_1,\, z_1)^2,$$

that is,

$$(1,\, 1,\, 1,\, \lambda,\, \mu,\, \nu \mid x,\, y,\, z)^2 = (1,\, 1,\, 1,\, \lambda_1,\, \mu_1,\, \nu_1 \mid x_1,\, y_1,\, z_1)^2,$$

or the linear relations between $(x,\, y,\, z)$ and $(x_1,\, y_1,\, z_1)$ are such as to transform one of these quadric functions into the other: the two quadrics, in fact, denote the squared distance from the origin expressed in terms of the coordinates $(x,\, y,\, z)$ and $(x_1,\, y_1,\, z_1)$ respectively.

Since the nine cosines are connected by six equations, there should exist values containing three arbitrary constants, and satisfying these equations identically: but, by what just precedes, it appears that the problem of determining these values is, in fact, that of finding the linear transformation between two given quadric functions: the problem of the linear transformation of a quadric function into itself has an elegant solution; but it would seem that this is not the case for the transformation between two different functions.

The foregoing equation

$$K = (a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma)^2$$

is a relation between λ, μ, ν, the cosines of the sides of a spherical triangle, and $(\alpha,\, \beta,\, \gamma)$ the cosines of the distances of a point P from the three vertices: it can be at once verified by means of the relation $A + B + C = 2\pi$, and thence

$$1 - \cos^2 A - \cos^2 B - \cos^2 C + 2\cos A \cos B \cos C = 0,$$

which connects the angles A, B, C which the sides subtend at P. Writing a, b, c for λ, μ, ν, and f, g, h for α, β, γ, the relation is

$$\begin{aligned} K &= (a,\, b,\, c,\, f,\, g,\, h \mid f,\, g,\, h)^2, \\ &= af^2 + bg^2 + ch^2 + 2(bc - a)gh + 2(ca - b)hf + 2(ab - c)fg, \end{aligned}$$

viz. this is

$$1 - a^2 - b^2 - c^2 - f^2 - g^2 - h^2 + 2abc + 2agh + 2bhf + 2cfg - a^2f^2 - b^2g^2 - c^2h^2 + 2bcgh + 2cahf + 2abfg = 0;$$

where $(a,\, b,\, c,\, f,\, g,\, h)$ are the cosines of the sides of a spherical quadrangle: (a, b, c), (a, h, g), (h, b, f), (g, f, c) belong respectively to sides forming a triangle, and the remaining sides (f, g, h), (b, c, f), (c, a, g), (a, b, h) are sides meeting in a vertex.

The equation

$$K\nu_1 = (a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma \mid \alpha',\, \beta',\, \gamma')$$

is a relation between λ, μ, ν, the cosines of the sides of a spherical triangle; α, β, γ, the cosines of the distances of a point P from the three vertices; α', β', γ', the cosines of the distances of a point Q from the three vertices; and ν_1, the cosine of the distance PQ.

Drawing a figure, it is at once seen that

$$\nu_1 = \alpha\alpha' + \sqrt{1 - \alpha^2}\,\sqrt{1 - \alpha'^2}\,\cos(\theta - \theta'),$$

where

$$\cos\theta = \frac{\beta - \alpha\nu}{\sqrt{1 - \alpha^2}\sqrt{1 - \nu^2}},$$

and therefore

$$\sin\theta = \frac{\sqrt{V}}{\sqrt{1 - \alpha^2}\sqrt{1 - \nu^2}};$$

also

$$\cos\theta' = \frac{\beta' - \alpha'\nu}{\sqrt{1 - \alpha'^2}\sqrt{1 - \nu^2}},$$

and therefore

$$\sin\theta' = \frac{\sqrt{V'}}{\sqrt{1-\alpha'^2}\sqrt{1-\nu^2}},$$

the values of V, V' being

$$V = 1-\alpha^2-\beta^2-\nu^2+2\alpha\beta\nu, \qquad V' = 1-\alpha'^2-\beta'^2-\nu^2+2\alpha'\beta'\nu;$$

the resulting value of ν_1 is therefore

$$\nu_1 = \alpha\alpha' + \frac{1}{1-\nu^2}\{(\beta-\alpha\nu)(\beta'-\alpha'\nu)+\sqrt{V}\sqrt{V'}\}.$$

The equations

$$K = (a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma)^2, \qquad K = (a,\, \dots \mid \alpha',\, \beta',\, \gamma')^2$$

give

$$\nabla V = -K^2, \qquad \nabla V' = -K^2,$$

and we therefore have

$$(g\alpha+f\beta+c\gamma \mid g\alpha'+f\beta'+c\gamma') = -\frac{1}{c}\sqrt{V}\sqrt{V'};$$

recollecting that $1-\nu^2 = c$, the formula thus is

$$\nu_1 = \alpha\alpha' + \frac{1}{c}\{(\beta-\alpha\nu)(\beta'-\alpha'\nu) + (g\alpha+f\beta+c\gamma \mid g\alpha'+f\beta'+c\gamma')\},$$

or say,

$$K\nu_1 = K\alpha\alpha' + \frac{1}{c}\{c^2\,(\beta-\alpha\nu \mid \beta'-\alpha'\nu) + (g\alpha+f\beta \mid g\alpha'+f\beta')\} + g(\alpha\gamma'+\alpha'\gamma) + f(\beta\gamma'+\beta'\gamma) + c\gamma\gamma'.$$

The sum of the first and second terms is readily found to be

$$= a\alpha\alpha' + b\beta\beta' + h(\alpha\beta'+\alpha'\beta),$$

and the equation thus becomes

$$K\nu_1 = (a,\, b,\, c,\, f,\, g,\, h \mid \alpha,\, \beta,\, \gamma \mid \alpha',\, \beta',\, \gamma'),$$

as it should do.

743.

ON THE NEWTON-FOURIER IMAGINARY PROBLEM.

[From the Proceedings of the Cambridge Philosophical Society, vol. iii. (1880), pp. 231, 232.]

The Newtonian process of approximation to the root of a numerical equation $f(u) = 0$, consists in deriving from an assumed approximate root ξ a new value $\xi_1 = \xi - \frac{f(\xi)}{f'(\xi)}$, which should be a closer approximation to the root sought for: taking the coefficients of $f(u)$ to be real, and also the root sought for, and the assumed value ξ, to be each of them real, Fourier investigated the conditions under which ξ_1 is in fact a closer approximation. But the question may be looked at in a more general manner: ξ may be any real or imaginary value, and we have to inquire in what cases the series of derived values

$$\xi_1 = \xi - \frac{f(\xi)}{f'(\xi)}, \quad \xi_2 = \xi_1 - \frac{f(\xi_1)}{f'(\xi_1)}, \; \dots$$

converge to a root, real or imaginary, of the equation $f(u) = 0$. Representing as usual the imaginary value ξ, $= x+iy$, by means of the point whose coordinates are x, y, and in like manner ξ_1, $= x_1+iy_1$, &c.,

then we have a problem relating to an infinite plane; the roots of the equation are represented by points A, B, $C, \ldots$; the value ξ is represented by an arbitrary point P; and from this by a determinate geometrical construction we obtain the point P_1, and thence in like manner the points P_2, $P_3, \ldots$ which represent the values ξ_1, ξ_2, $\xi_3, \ldots$ respectively. And the problem is to divide the plane into regions, such that, starting with a point P, anywhere in one region, we arrive ultimately at the root A; anywhere in another region we arrive ultimately at the root B; and so on for the several roots of the equation. The division into regions is made without difficulty in the case of a quadric equation: but in the next succeeding case, that of a cubic equation, it is anything but obvious what the division is: and the author had not succeeded in finding it.

744.

TABLE OF $\Delta^m 0^n \div \Pi(m)$ UP TO $m = n = 20$.

[From the Transactions of the Cambridge Philosophical Society, vol. xiii. Part i. (1881), pp. 1–4. Read October 27, 1879.]

The differences of the powers of zero, $\Delta^m 0^n$, present themselves in the Calculus of Finite Differences, and especially in the applications of Herschel's theorem,

$$f(e^x) = f(1+\Delta)\, e^{0x},$$

for the expansion of the function of an exponential. A small Table up to $\Delta^5 0^n$ is given in Herschel's *Examples* (Camb. 1820), and is reproduced in the treatise on Finite Differences (1843) in the Encyclopaedia Metropolitana. But, as is known, the successive differences $\Delta 0^n$, $\Delta^2 0^n$, $\Delta^3 0^n$, $\ldots$ are divisible by 1, $1 \cdot 2$, $1{\cdot}2{\cdot}3, \ldots$ and generally $\Delta^m 0^n$ is divisible by $1{\cdot}2{\cdot}3 \cdots m, = \Pi(m)$; these quotients are much smaller numbers, and it is therefore desirable to tabulate them rather than the undivided differences $\Delta^m 0^n$: moreover, it is easier to calculate them. A table of the quotients $\Delta^m 0^n \div \Pi(m)$, up to $m = n = 12$ is in fact given by Grünert, *Crelle*, t. xxv. (1843), p. 279, but without any explanation in the heading of the meaning of the tabulated numbers $C_n^m, = \Delta^m 0^n \div \Pi(n)$, and without using for their determination the convenient formula $C_n^{m+1} = nC_n^m + C_{n-1}^m$ given by Björling in a paper, *Crelle*, t. xxviii. (1844), p. 284. The formula in question, say

$$\frac{\Delta^m 0^n}{\Pi(m)} = m\,\frac{\Delta^m 0^{n-1}}{\Pi(m)} + \frac{\Delta^{m-1} 0^{n-1}}{\Pi(m-1)},$$

is given in the second edition (by Moulton) of Boole's *Calculus of Finite Differences*, (London, 1872), p. 28, under the form

$$\Delta^m 0^n = m\,(\Delta^{m-1} 0^{n-1} + \Delta^m 0^{n-1}).$$

It occurred to me that it would be desirable to extend the table of the quotients $\Delta^m 0^n \div \Pi(m)$, up to $m = n = 20$. The calculation is effected very readily by means of the foregoing theorem, which is used in the following form; viz. any column of the table for instance the fifth, being

$$\begin{array}{lcl}
A, & \text{then the following column is} & A, \\
B, & \ldots & 2B + A, \\
C, & \ldots & 3C + B, \\
D, & \ldots & 4D + C, \\
E, & \ldots & 5E + D, \\
 & & +E;
\end{array}$$

and then we obtain a good verification by taking the sum of the terms in the new column, and comparing it with the value as calculated from the formula.

$$\text{Sum} = 2A + 3B + 4C + 5D + 6E.$$

Observe that, in the two calculations, we take successive multiples such as $4D$ and $5D$ of each term of the preceding column, and that the verification is thus a safe- guard against any error of multiplication or addition.

Table, No. 1, of $\Delta^m 0^n \div \Pi(m)$.

n/m	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	1													
2	1	1												
3	1	3	1											
4	1	7	6	1										
5	1	15	25	10	1									
6	1	31	90	65	15	1								
7	1	63	301	350	140	21	1							
8	1	127	966	1 701	1 050	266	28	1						
9	1	255	3 025	7 770	6 951	2 646	462	36	1					
10	1	511	9 330	34 105	42 525	22 827	5 880	750	45	1				
11	1	1 023	28 501	145 750	246 730	179 487	63 987	11 880	1 155	55	1			
12	1	2 047	86 526	611 501	1 379 400	1 323 652	627 396	159 027	22 275	1 705	66	1		
13	1	4 095	261 625	2 532 530	7 508 501	9 321 312	5 715 424	1 899 612	359 502	39 325	2 431	78	1	
14	1	8 191	788 970	10 391 745	40 075 035	63 436 373	49 329 280	20 912 320	5 135 130	752 752	66 066	3 367	91	1

(Continuation of Table No. 1 for $m = 8, \dots, 13$ and $n = 2, \dots, 20$.)

n/m	8	9	10	11	12	13
2	16 383	32 767	65 535	131 071	262 143	524 287
3	2 375 101	7 141 686	21 457 825	64 439 010	193 448 101	580 606 446
4	48 355 950	171 798 901	694 337 290	2 798 806 985	11 259 666 950	45 232 116 901
5	210 766 920	1 096 190 550	5 652 751 651	28 958 095 545	147 589 284 710	749 206 090 500
6	420 693 273	2 734 926 558	17 505 749 898	110 687 251 039	693 081 601 779	4 306 078 895 384
7	408 741 333	3 281 882 604	25 708 104 786	197 462 483 400	1 492 924 634 839	11 143 554 046 652
8	216 627 840	2 141 764 053	20 415 995 028	189 036 065 010	1 709 751 003 480	15 170 932 662 679
9	67 128 490	820 784 250	9 528 822 303	106 175 396 755	1 144 614 626 805	12 011 282 644 725
10	12 662 650	193 754 990	2 758 334 150	37 112 163 803	477 297 033 785	5 917 584 964 665
11	1 479 478	28 936 908	512 060 978	8 391 004 908	129 413 217 791	1 900 842 429 486
12	106 470	2 757 118	62 022 324	1 256 328 866	23 466 951 300	411 016 633 391
13	4 550	166 620	4 910 178	125 864 638	2 892 439 160	61 068 660 380
14	105	6 020	249 900	8 408 778	243 577 530	6 302 524 580
15	1	120	7 820	367 200	13 916 778	452 329 200
16		1	136	9 996	527 136	22 350 954
17			1	153	12 597	741 285
18				1	171	15 675
19					1	190
20						1

Writing down the sloping lines as columns thus:

1	2	3 − 4	5	6	7	8 etc.
(0)	(2)	(4) (6)	(8) (10)	(12)		(14) etc.

Table, No. 2 (explanation infrà).

	1	2	3	4	5	6	7
0	1	1	1	1	1	1	1
1		2	6	14	30	62	126
2		1	12	61	240	841	2 772
3			10	124	890	5 060	25 410
4			3	131	1 830	16 990	127 953
5				70	2 226	35 216	401 436
6				15	1 600	47 062	836 976
7					630	40 796	1 196 532
8					105	21 225	1 182 195
9						10 930	795 718
10						945	349 020
11							90 090
12							10 395

We have, by means of this Table, the general expressions of $\Delta^r 0^r$, $\Delta^{r-1}0^r$, $\Delta^{r-2}0^r, \ldots$ up to $\Delta^{r-6}0^r$, viz. the formulae are

$$\begin{aligned}
\Delta^r 0^r \div \Pi(r) &= 1,\\
\Delta^{r-1}0^r \div \Pi(r-1) &= 1 + 2\binom{r-1}{1} + 1\binom{r-1}{2},\\
\Delta^{r-2}0^r \div \Pi(r-2) &= 1 + 6\binom{r-2}{1} + 12\binom{r-2}{2} + 10\binom{r-2}{3} + 3\binom{r-2}{4},\\
&\text{\&c., \&c.,}
\end{aligned}$$

where the numerical coefficients are the numbers in the successive columns of the table; and where for shortness $\binom{r-m}{k}$ is written to denote the binomial coefficient $\frac{[r-m]^k}{[k]^k}$. For instance, $r = 10$, we have

$$\Delta^8 0^{10} \div \Pi(8) = 1 + 6\cdot 7 + 12\cdot 21 + 10\cdot 35 + 3\cdot 35, \ = 750,$$

agreeing with the principal Table. It will be observed that, in the successive columns of the Table, the last terms are 1, 1, $1\cdot 3$, $1\cdot 3\cdot 5$, $1\cdot 3\cdot 5\cdot 7$, $1\cdot 3\cdot 5\cdot 7\cdot 9$, and $1\cdot 3\cdot 5\cdot 7\cdot 9\cdot 11$. This is itself a good verification: I further verified the last column by calculating from it the value of $\Delta^{14}0^{20} \div \Pi(14)$, $= 6\,302\,524\,580$ as above. The Table shows that we have $\Delta^{r-m}0^r \div \Pi(r-m)$ given as an algebraical rational and integral function of r, of the degree $2m$. But the terms from the top of a column, $\Delta 0^r = 1$, $\Delta^2 0^r \div 1\cdot 2 = 2^{r-1} - 1$, &c., are not algebraical functions of r.

22 October, 1879.

745.

ON THE SCHWARZIAN DERIVATIVE, AND THE POLYHEDRAL FUNCTIONS.

[From the Transactions of the Cambridge Philosophical Society, vol. xiii. Part i. (1881), pp. 5–68. Read March 8, 1880.]

The quotient s of any two solutions of a linear partial differential equation of the second order, $\frac{d^2v}{dx^2} + p\frac{dv}{dx} + qv = 0$, is determined by a differential equation of the third order

$$\frac{\dfrac{d^3s}{dx^3}}{\dfrac{ds}{dx}} - \frac{3}{2}\left(\frac{\dfrac{d^2s}{dx^2}}{\dfrac{ds}{dx}}\right)^2 = -\tfrac{1}{2}\left(p^2 + 2\frac{dp}{dx} - 4q\right),$$

where the function on the left-hand is what I call the Schwarzian Derivative; or say this derivative is

$$\{s,\, x\} = \frac{s'''}{s'} - \tfrac{3}{2}\left(\frac{s''}{s'}\right)^2,$$

where the accents denote differentiations in regard to the second variable x of the symbol.

Writing in general $(a,\, b,\, c\, ..\mid X,\, Y,\, Z)^2$ to denote a quadric function

$$(a,\, b,\, c,\, \tfrac{1}{2}(a-b-c),\, \tfrac{1}{2}(-a+b-c),\, \tfrac{1}{2}(-a-b+c) \mid X,\, Y,\, Z)^2,$$

then, if the equation of the second order be that of the hypergeometric series, generalised by a homographic transformation upon the variable x, the resulting differ- ential equation of the third order is of the form

$$\{s,\, x\} = (a,\, b,\, c\, ..)\left(\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2,$$

and, presenting themselves in connexion with the algebraically integrable cases of this equation, we have rational and integral functions of s, derived from the polygon, the double pyramid, and the five regular solids. They are called Polyhedral Functions.

The Schwarzian Derivative occurs implicitly in Jacobi's differential equation of the third order for the modulus in the transformation of an elliptic function (*Fund. Nova*, 1829, p. 79, [*Ges. Werke*, t. i., p. 133]) and in Kummer's fundamental equation for the transformation of a hypergeometric series (Kummer, 1836: see list of Memoirs): but it was first explicitly considered and brought into notice in the two Memoirs of Schwarz[1], 1869 and 1873. The latter of these, relating to the algebraic integration of the differential equation for the hypergeometric series, is the fundamental Memoir upon the subject, but the theory is in some material points completed in the Memoirs by Klein and Brioschi.

The following list of Memoirs, relating as well to the Polyhedral Functions as to the Schwarzian Derivative, is arranged nearly in chronological order.

KUMMER, Ueber die hypergeometrische Reihe $1+\dfrac{\alpha\cdot\beta}{1\cdot\gamma}x+\ldots$ *Crelle*, t. xv. (1836), pp. 39–83 and 127–172.

SCHWARZ, Ueber einige Abbildungsaufgaben. *Crelle-Borchardt*, t. lxx. (1869), pp. 105–120.

Ueber diejenigen Fälle in welchen die Gaussische hypergeometrische Reihe eine algebraische Function ihres vierten Elementes darstellt. *Do.* t. lxxv. (1873), pp. 292–335.

CAYLEY, Notes on Polyhedra. *Quart. Math. Jour.* t. vii. (1866), pp. 304–316; [375].

On the Regular Solids. *Do.* t. xv. (1878), pp. 127–131; [679].

FUCHS, Ueber diejenigen Differentialgleichungen zweiter Ordnung welche algebraische Integralen besitzen, und eine Anwendung der Invariantentheorie. *Crelle-Borchardt*, t. lxxxi. (1875), pp. 97–142.

KLEIN, Ueber binäre Formen mit linearen Transformationen in sich selbst. *Math. Ann.* t. ix. (1875), pp. 183–209.

BRIOSCHI, Extrait d'une lettre à M. Klein. *Math. Ann.* t. xi. (1877), pp. 111–114.

KLEIN, Ueber lineare Differentialgleichungen. *Math. Ann.* t. xi. (1877), pp. 115–118.

BRIOSCHI, La théorie des formes dans l'intégration des équations différentielles linéaires du second ordre. *Math. Ann.* t. xi. (1877), pp. 401–411.

GORDAN, Ueber endliche Gruppen linearer Transformationen einer Veranderlichen. *Math. Ann.* t. xii. (1877), pp. 23–46.

Binäre Formen mit verschwindenden Covarianten. *Math. Ann.* t. xii. (1877), pp. 147–166.

KLEIN, Ueber lineare Differentialgleichungen. *Math. Ann.* t. xii. (1877), pp. 167–179.

Weitere Untersuchungen über das Icosaeder. *Math. Ann.* t. xii. (1877), pp. 503–560.

CAYLEY, On the Correspondence of Homographies and Rotations. *Math. Ann.* t. xv. (1879), pp. 238–240; [660].

On the finite Groups of linear transformations of a Variable. *Math. Ann.* t. xvi. (1880), pp. 260–263, and pp. 439–440; [752].

I propose in the present Memoir to consider the whole theory: and, in particular, to give some additional developments in regard to the Polyhedral Functions.

[1]Schwarz, *Ges. Werke*, t. ii., p. 351, remarks that the Derivative occurs implicitly in a memoir by Lagrange, "Sur la construction des cartes géographiques," (1779), Œuvres, t. iv., p. 651.

I remark that Schwarz starts with the foregoing differential equation of the third order

$$\{s,\, x\} = (a,\, b,\, c\, ..)\left(\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2,$$

and he shows (by very refined reasoning founded on the theory of conformable figures, which will be in part reproduced) that this equation is, in fact, algebraically integrable for 16 different sets of values of the coefficients a, b, c. It may I think be taken to be part of his theory, although not very clearly brought out by him, that these integrals are some of them of the form, $x =$ rational function of s: others of the form, rational function of $x =$ rational function of s; the rational functions of s being in fact the same in the last as in the first set of solutions: they are quotients of Polyhedral functions.

But as regards the second set of cases, the solution of these, introducing for con- venience a new variable z in place of s, may be made to depend upon the solution in the form, $x =$ rational function of z, of an equation of a somewhat similar form, but involving two quadric functions of x and z respectively, viz. the equation

$$\{x,\, z\} - (a,\, b,\, c\, ..)\left(\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = (\alpha,\, \beta,\, \gamma\, ..)\left(\frac{1}{z-\alpha},\, \frac{1}{z-\beta},\, \frac{1}{z-\gamma}\right)^2,$$

and we have the theorem that the solution of this equation depends upon the determ- ination of P, Q, R rational and integral functions of z, containing each of them multiple factors, which are such that $P+Q+R=0$. Using accents to denote differ- entiation in regard to z, this implies $P'+Q'+R'=0$, and consequently

$$QR' - Q'R = RP' - R'P = PQ' - P'Q.$$

Further, they are such that the equal functions $QR'-Q'R$, $RP'-R'P$, $PQ'-P'Q$ contain only factors which are factors of P, Q or R.

In fact, writing $f,\, g,\, h = b-c,\, c-a,\, a-b$, the required relation between x, z is then expressed in the symmetrical form $f(x-a) : g(x-b) : h(x-c) = P : Q : R$.

The last-mentioned differential equation is considered by Klein and Brioschi: the solutions in 13 cases, or such of them as had not been given by Schwarz, were obtained by Brioschi: and those of the remaining 3 cases, subject to a correction in one of them, were afterwards obtained by Klein.

The first part of the present Memoir relates, say to the foregoing equation

$$\{s,\, x\} = (a,\, b,\, c\, ..)\left(\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2,$$

although the other form in $\{x,\, z\}$ may equally well be regarded as the fundamental form.

We consider in the theory:

A. The Derivative $\{s,\, x\}$, meaning as above explained.

B. Quadric functions of any three or more inverts $\dfrac{1}{x-l}$.

C. Rational and integral functions P, Q, R having a sum $= 0$, and which are such that $QR'-Q'R$, $= RP'-R'P$, $= PQ'-P'Q$, contains only the factors of P, Q, R.

D. The differential equation of the third order.

E. The Schwarzian theory in regard to conformable figures and the corresponding values of the imaginary variables s and x.

F. Connexion with the differential equation for the hypergeometric series.

The second part of the Memoir relates to the Polyhedral Functions.

The paragraphs of the whole Memoir are numbered consecutively.

Part I.

The Derivative $\{s,\, x\}$. Art. Nos. 1 to 7.

2. The derivative $\{s,\, x\}$ may be transformed in regard to either or both of the variables.

Suppose, first, that s is a function of the new variable S, (hence also S is a function of x): using subscript numbers to denote differentiations in regard to S, and the accents as before for differentiations in regard to x, we have

$$s' = S' s_1,$$

whence, differentiating the logarithms,

$$\frac{s''}{s'} = S' \, \frac{s_2}{s_1} + \frac{S''}{S'},$$